

MEADOW LAKE, CO, USA

WMO: 720852

Lat: 38.946N

Lon: 104.570W

Elev: 6874

StdP: 11.39

Time Zone: -7.00 (NAM)

Period: 10-19

WBAN: 00280

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	0.8	5.9	-10.9	3.9	25.8	-7.7	4.7	27.6	30.9	39.0	27.3	38.1	7.8	350	0.621

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	26.3	90.0	56.0	87.1	56.0	84.4	55.6	61.6	74.8	60.4	74.5	59.4	74.2	10.9	150

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
58.4	94.4	63.4	56.9	89.3	62.8	55.3	84.2	62.5	31.1	74.5	30.2	74.4	29.3	73.9	66.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 26.9	23.6	19.9	DB	-7.4	94.2	6.3	2.2	-11.9	95.8	-15.6	97.1	-19.2	98.4	-23.8	100.0	(4)
(5)			WB	-7.3	64.6	6.2	1.3	-11.8	65.5	-15.4	66.3	-18.9	67.0	-23.4	67.9	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	48.7	31.1	30.8	39.9	44.8	52.8	66.5	70.1	67.8	62.1	47.7	39.0	30.5		
	DBStd	16.73	9.66	11.87	9.82	8.50	8.58	6.16	4.47	4.40	6.75	10.03	9.22	10.00		
	HDD50	2827	587	540	330	193	75	1	0	0	5	155	338	603		
	HDD65	6375	1052	958	777	606	380	54	12	25	128	535	780	1068		
	CDD50	2351	0	3	18	37	162	495	623	551	368	85	9	0		
	CDD65	423	0	0	0	0	3	98	170	111	41	0	0	0		
	CDH74	5535	0	0	2	6	159	1520	1885	1242	679	42	0	0		
	CDH80	1842	0	0	0	0	22	605	715	344	153	3	0	0		
(14) Wind	WSAvg	9.0	8.5	9.7	9.9	10.9	10.2	9.3	8.2	7.7	8.2	8.8	8.8	8.4		
(15) Precipitation	PrecAvg	17.8	0.4	0.4	0.9	1.9	2.5	2.0	3.3	3.1	1.4	0.9	0.6	0.4		
	PrecMax	27.5	0.9	1.3	2.6	7.5	9.0	5.4	6.2	6.0	4.8	2.8	2.6	1.5		
	PrecMin	8.2	0.0	0.0	0.0	0.0	0.9	0.4	0.7	0.1	0.0	0.0	0.0	0.1		
	PrecStd	4.6	0.2	0.3	0.7	1.5	1.9	1.4	1.4	1.4	1.1	0.7	0.5	0.3		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	62.8	66.0	72.4	75.3	82.9	94.7	92.3	89.5	87.8	78.7	69.1	61.0		
		MCWB	40.3	42.2	44.8	46.7	50.7	55.4	57.0	56.1	54.2	49.8	43.3	39.5		
	2%	DB	56.7	59.4	68.0	70.5	78.8	90.6	89.9	86.3	84.3	74.2	64.3	56.8		
		MCWB	37.9	38.8	43.0	45.3	49.3	54.7	57.3	56.1	53.8	47.6	41.8	37.4		
	5%	DB	52.4	55.0	63.4	67.0	74.7	86.8	87.1	84.1	81.6	71.1	60.5	52.5		
		MCWB	35.9	37.1	40.7	44.0	48.5	54.0	57.5	56.5	53.0	46.3	40.6	36.0		
	10%	DB	47.8	50.2	58.8	63.3	70.6	83.0	84.3	81.4	78.4	66.4	55.7	47.9		
		MCWB	33.7	34.9	39.2	42.5	48.0	53.5	57.3	56.4	52.5	45.1	37.8	33.6		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	41.5	42.3	45.5	49.3	54.3	62.1	63.2	63.6	60.5	53.1	44.3	40.5		
		MCDB	59.6	64.5	70.6	61.8	67.3	75.3	74.6	75.6	73.4	66.2	63.6	58.3		
	2%	WB	38.3	39.4	43.6	47.2	52.3	59.8	61.8	61.4	58.2	50.5	42.9	38.1		
		MCDB	54.9	57.2	65.4	64.9	66.9	74.7	75.2	73.8	71.0	66.6	63.1	55.2		
	5%	WB	36.4	37.5	41.7	45.6	51.0	58.1	60.7	60.0	56.5	48.2	41.0	36.3		
		MCDB	51.7	54.4	59.9	63.6	66.7	73.5	75.0	74.0	70.2	65.2	59.0	51.7		
	10%	WB	33.8	35.1	39.9	43.7	49.8	56.7	59.8	58.8	55.0	46.4	38.9	33.9		
		MCDB	46.9	49.6	57.0	60.9	65.9	73.3	74.6	73.2	70.2	63.6	54.4	46.7		
(35) Mean Daily Temperature Range	5% DB	MDBR	25.1	24.4	26.5	25.6	24.5	28.5	26.3	25.6	26.1	26.4	25.6	24.9		
		MCDBR	30.3	30.3	32.5	33.2	31.8	34.4	30.8	29.6	30.9	33.3	31.2	32.1		
		MCWBR	16.0	15.0	14.8	13.9	11.3	10.2	8.0	8.4	9.7	13.2	14.4	17.2		
	5% WB	MCDBR	29.5	30.0	30.8	30.3	28.3	27.4	25.2	24.1	24.4	29.0	30.0	30.7		
		MCWBR	15.9	15.2	14.7	13.2	10.3	10.5	8.3	8.8	8.9	12.1	14.3	16.9		
(40) Clear-Sky Solar Irradiance	taub		0.225	0.230	0.250	0.275	0.293	0.305	0.331	0.321	0.287	0.254	0.234	0.226		
	taud		2.645	2.660	2.601	2.535	2.546	2.529	2.501	2.520	2.588	2.667	2.642	2.633		
	Ebn at Noon		315	326	325	319	312	307	298	300	306	311	308	307		
	Edh at Noon		21	23	28	32	32	33	33	32	28	23	21	19		
(44) All-Sky Solar Radiation	RadAvg		848	1143	1542	1876	2085	2330	2225	2017	1726	1287	937	752		
	RadStd		43	72	112	117	156	101	91	63	85	75	52	40		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (28 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+126	+59

Nomenclature: See separate page